Paramount Diagnostic Lab

D-Dimer

Test Details

METHODOLOGY

Immunoturbidimetric assay

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

The Innovance® D-dimer assay is intended for use in conjunction with a nonhigh clinical pretest probability (PTP) assessment model to exclude deep vein thrombosis (DVT) and pulmonary embolism (PE).[6] This test can be used to exclude VTE with nonhigh pretest probability (ie, low or low/moderate pretest probability). In an exclusion strategy, a D-dimer below the established threshold in a nonhigh pretest probability patient does not require further testing to exclude VTE.

Special Instructions

If the patient's hematocrit exceeds 55%, the volume of citrate in the collection tube must be adjusted. Refer to Coagulation Collection Procedures (/node/455) for directions.

Limitations

Results of this test should always be interpreted in conjunction with the patient's medical history, clinical presentation, and other findings. DVT clinical diagnosis should not be based on the result of Innovance® D-dimer alone.

D-dimer levels can be elevated in many clinical circumstances, especially in hospitalized patients. D-dimer should not be used as an aid for exclusion of venous thrombosis or pulmonary embolism in pediatric patients in any circumstance and adult patients with[6-8]:

• Therapeutic dose anticoagulant administered for >24 hours before D-dimer is measured

• Thrombosis distal to the knee only

• Fibrinolytic therapy within previous seven days

• Upper extremity thrombosis

• D-dimer levels may be falsely negative if the elapsed time between thrombosis onset and D-dimer measurement is sufficient such that D-dimer has been cleared from the circulation.

The following conditions are associated with an increase in D-dimer concentrations, even in the absence of venous thrombosis:

• Aortic aneurysm

• Trauma or surgery within previous four weeks

• Disseminated malignancies

• Disseminated intravascular coagulation

• Sickle cell disease

• Sepsis, severe infections, pneumonia, severe skin infections

• Liver cirrhosis

• Pregnancy

Also note[6]:

• Patient samples may contain heterophilic antibodies (eg, human antimouse antibodies [HAMA] and rheumatoid factors) that could react in immunoassays to yield falsely elevated results. This assay has been designed to minimize interference from heterophilic antibodies. Nevertheless, complete elimination of this interference from all patient specimens cannot be guaranteed.

• Patients with subsegmental/peripheral PE or distal DVT may have a normal Innovance® D-dimer result.[9,10]

• Exclusionary claim of PE in patients with high PTP scores has not been established.

Custom Additional Information

Coagulation activation results in the cleavage of fibrinogen to fibrin monomer.[7,8] The fibrin monomers spontaneously aggregate to fibrin and are cross-linked by factor XIII; this produces a fibrin clot. In response to the coagulation process the fibrinolytic system is activated resulting in the conversion of plasminogen into plasmin, which cleaves fibrin (and fibrinogen) into the fragments D and E. Due to cross-linkage between D-domains in the fibrin clot, the action of plasmin releases fibrin degradation products with cross-linked D-domains. The smallest unit is D-dimer. Detection of D-dimers, which specifies cross-linked fibrin degradation products generated by reactive fibrinolysis, is an indicator of coagulation activity. Fibrin degradation products are not consistently "D-dimer" but are a mixture of fragments and complexes of different molecular weight. The presence of D-dimer confirms that both thrombin and plasmin have been generated since it can only be produced as the result of the plasmin degradation of cross-linked fibrin. The in vivo half-life of D-dimer is approximately eight hours.[11]

Elevated D-dimer levels are observed in all diseases and conditions with increased coagulation activation, eg, thromboembolic disease, DIC, acute aortic dissection, myocardial infarction, malignant diseases, obstetrical complications, third trimester of pregnancy, surgery, or polytrauma.[12-17] However, in the context of venous thromboembolism, symptoms being present since a certain period of time, eg, longer than a week, may produce normal D-dimer values.[18] For the diagnosis of DIC a scoring system has been suggested, in which elevated D-dimer levels represent the major indicator of DIC.[12] While increased levels of D-dimer are not specific for DVT or PE, low D-dimer levels may be used to rule out these conditions. The negative predictive values for DVT and PE are approaching 100% for the Innovance D-dimer assay employing a cutoff of <0.5 mg/L FEU. The negative predictive value is further enhanced through the use of a clinical probability model along with D-dimer in the decision process.[12-15] Values less than 0.5 mg/L FEU in an individual with a low clinical risk of venous thrombosis can serve as the basis for not performing more expensive diagnostic tests for DVT and PE. Patients with results greater than this cutoff require further diagnostic testing to establish the diagnosis.

D-dimer levels are known to increase with age; eg, median D-dimer levels in apparently healthy men aged 75 to 79 are about twice as high as in men aged 60 to 64.[19] This increase in the basal D-dimer concentration is responsible for the decrease in the specificity of D-dimer measurements for exclusion of VTE in the elderly. Several studies on D-dimer for exclusion of venous thromboembolism (VTE) have been reëvaluated using an age-specific cutoff.[20-23] The aim was to improve the effectiveness of D-dimer testing in ruling out VTE.[20-23] The cutoff provided with the patient result is the manufacturer-determined value for exclusion of VTE; however, it has been determined that D-dimer values increase with age, and this can make VTE exclusion of an older population difficult. To address this, the American College of Physicians, based on best available evidence and recent guidelines, recommends that clinicians use age-adjusted D-dimer thresholds in patients greater than 50 years of age with: **(a)** a low probability of PE who do not meet all "pulmonary-embolism-rule-out criteria," or **(b)** in those with intermediate probability of PE.[21] The formula for an age-adjusted D-dimer cutoff is "age/100." For example, a 60-year-old patient would have an age-adjusted cutoff of 0.6 mg/L FEU, and an 80-year-old patient would have an age-adjusted cutoff of 0.8 mg/L FEU.

Specimen Requirements

Specimen

Plasma, **frozen**

Volume

2 mL

Minimum Volume

1 mL

Container

Blue-top (sodium citrate) tube

Collection Instructions

Citrated plasma samples should be collected by double centrifugation. Blood should be collected in a blue-top tube containing 3.2% buffered sodium citrate.[1] Evacuated collection tubes must be filled to completion to ensure a proper blood to anticoagulant ratio.[2,3] The sample should be mixed immediately by gentle inversion at least six times to ensure adequate mixing of the anticoagulant with the blood. A discard tube is not required prior to collection of coagulation samples.[4,5] When noncitrate tubes are collected for other tests, collect sterile and nonadditive (red-top) tubes prior to citrate (blue-top) tubes. Any tube containing an alternate anticoagulant should be collected after the blue-top tube. Gel-barrier tubes and serum tubes with clot initiators should also be collected after the citrate tubes. Centrifuge for 10 minutes and carefully remove $^{2}/_{3}$ of the plasma using a plastic transfer pipette, being careful not to disturb the cells. Deliver to a plastic transport tube, cap, and recentrifuge for 10 minutes. Transfer the plasma into a Labcorp PP transpak frozen purple tube with screw cap (Labcorp No. 49482). Freeze plasma immediately and maintain frozen until tested. To avoid delays in turnaround time when requesting multiple tests on frozen samples, **please submit separate frozen specimens for each test requested**.

Please print and use the *Volume Guide for Coagulation Testing* *(/tests/related-documents/L5901)* to ensure proper draw volume.

Stability Requirements

Temperature	Period
Room temperature	Unacceptable
Refrigerated	Unacceptable
Frozen	4 weeks
Freeze/thaw cycles	Stable x1

Reference Range

0.00–0.49 mg/L FEU (Fibrinogen equivalent units)

Age (Years)	Age-Adjusted Cutoff (mg/L FEU)
50	0.00 – 0.50
51	0.00 – 0.51
52	0.00 – 0.52
53	0.00 – 0.53
54	0.00 – 0.54
55	0.00 – 0.55
56	0.00 – 0.56
57	0.00 – 0.57
58	0.00 – 0.58
59	0.00 – 0.59
60	0.00 – 0.60
61	0.00 – 0.61
62	0.00 – 0.62
63	0.00 – 0.63
64	0.00 – 0.64
65	0.00 – 0.65
66	0.00 – 0.66

Age (Years)	Age-Adjusted Cutoff (mg/L FEU)
67	0.00 – 0.67
68	0.00 – 0.68
69	0.00 – 0.69
70	0.00 – 0.70
71	0.00 – 0.71
72	0.00 – 0.72
73	0.00 – 0.73
74	0.00 – 0.74
75	0.00 – 0.75
76	0.00 – 0.76
77	0.00 – 0.77
78	0.00 – 0.78
79	0.00 – 0.79
80	0.00 – 0.80
81	0.00 – 0.81
82	0.00 – 0.82
83	0.00 – 0.83

Age (Years)	Age-Adjusted Cutoff (mg/L FEU)
84	0.00 – 0.84
85	0.00 – 0.85
86	0.00 – 0.86
87	0.00 – 0.87
88	0.00 – 0.88
89	0.00 – 0.89
90	0.00 – 0.90
91	0.00 – 0.91
92	0.00 – 0.92
93	0.00 – 0.93
94	0.00 – 0.94
95	0.00 – 0.95
96	0.00 – 0.96
97	0.00 – 0.97
98	0.00 – 0.98
99	0.00 – 0.99

Storage Instructions

Freeze.

Causes for Rejection

Gross hemolysis; clotted specimen; specimen thawed in transit; improper labeling

Footnotes

1. Adcock DM, Kressin DC, Marlar RA. Effect of 3.2% vs 3.8% sodium citrate concentration on routine coagulation testing. *Am J Clin Pathol.* 1997 Jan; 107(1):105-110. PubMed 8980376 (http://www.ncbi.nlm.nih.gov/pubmed/8980376)
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12. Wells PS. The role of qualitative D-dimer assays, clinical probability, and noninvasive imaging tests for the diagnosis of deep vein thrombosis and pulmonary embolism. *Semin Vasc Med.* 2005 Nov; 5(4):340-350. PubMed 16302155 (http://www.ncbi.nlm.nih.gov/pubmed/16302155)
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14. Taylor FB Jr, Toh CH, Hoots WK, Wada H, Levi M; Scientific Subcommittee on Disseminated Intravascular Coagulation (DIC) of the International Society on Thrombosis and Haemostasis (ISTH). Towards definition, clinical and laboratory criteria, and a scoring system for disseminated intravascular coagulation. *Thromb Haemost.* 2001 Nov; 86(5):1327-1330. PubMed 11816725 (http://www.ncbi.nlm.nih.gov/pubmed/11816725)
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